

C-DAC MUMBAI

Post Graduate Certificate Programme in Big Data Analytics (PGCP-BDA)

END-TERM EXAMINATION

Credit Risk Analytics & NPA Prediction

Subject	Credit Risk Analytics NPA Prediction & Modelling
Duration	
Maximum Marks	60
Pass Marks	
Date & Time	_____
Student Name	_____
Roll Number	_____
Batch / Section	_____

GENERAL INSTRUCTIONS

- This paper contains FIVE questions. Attempt ALL questions.
- Marks are indicated in brackets at the right of each sub-question.
- Answers must be supported with code, visualisations, statistical output, or quantified business rationale as specified.
- All Python notebooks must be submitted in executed form (.ipynb) with output cells visible.
- Clearly label every visualisation with a title, axis labels, and a minimum 3-sentence insight caption.
- Do NOT delete dirty or outlier records. Flag and impute as appropriate.
- Do NOT impute target variables (loan_status, lgd_pct). Drop rows with missing targets and log the count.
- Use random_state=42 for all train-test splits and cross-validation.

QUESTION 1 DATA ACQUISITION, JOINING & CLEANING**[30 Marks]**

The FinSight Bank dataset is distributed across nine CSV files linked by a common loan_id key. Your first task is to assemble and clean the full dataset before any analysis can begin.

- (a) Load all nine CSV files using `pd.read_csv()`. For files exceeding available RAM, use `chunksize` appropriately. Save the concatenated output as a Parquet file and report: (i) memory usage before and after downcasting, and (ii) file size of the resulting Parquet. (5)
- (b) Join all nine tables on `loan_id` using sequential left merges. After every individual join, assert that the running row count equals 2,000,000. Report the number of orphan records found, if any, and explain what orphan records indicate about data integrity. (6)
- (c) The dataset contains eight deliberately injected data quality issues across multiple columns. Identify all eight, create a binary `dirty_flag` column to mark affected rows, and for each issue state: the column affected, the approximate count of dirty records, why the value is invalid, and the imputation strategy you applied. (10)
- (d) Classify the missing-value pattern for each high-missing column (`mths_since_last_delinq`, `mort_acc`, `emp_length_years`, `il_util_pct`) as MCAR, MAR, or MNAR. Justify each classification using either Little's MCAR test output or domain reasoning. Apply the correct imputation strategy and verify with `.isnull().sum()` before and after. (6)
- (e) Apply winsorisation at the 1st and 99th percentile to the six most skewed numeric columns. Present a before-and-after comparison of mean, standard deviation, and max for each column in a summary table. (3)

QUESTION 2 EXPLORATORY DATA ANALYSIS**[25 Marks]**

All twelve visualisations below are compulsory. Each chart must include a descriptive title, correctly labelled axes, and a minimum 3-sentence insight caption explaining the business implication. Partial marks are deducted for missing captions or unlabelled axes.

- (a) Produce a bar chart and a pie chart showing the distribution of `loan_status`. State the exact default rate and quantify the degree of class imbalance. Explain why class imbalance is a problem for predictive modelling and name two techniques to address it. (3)
- (b) Plot overlapping KDE curves of `cibil_score` for defaulted and performing loans on the same axes. Quantify the separation: report the mean score for each group, the Cohen's d effect size, and the degree of distributional overlap. What does the overlap tell us about relying solely on CIBIL? (3)

- (c) Produce a 12-panel histogram grid covering all key numeric features. Identify which features exhibit right skew exceeding 2.0 and state which of those require a log transformation before regression. Apply the transformations and confirm skew reduction with `.skew()` values. (3)
- (d) Compute a Pearson correlation matrix for the top 20 numeric features and display it as an annotated heatmap. Identify any pair with $|r| > 0.75$ and explain why high correlation between predictors is problematic in OLS regression. (2)
- (e) Create six side-by-side boxplots (`int_rate`, `dti`, `cibil_score`, `annual_inc`, `revol_util`, `emp_length` vs. `loan_status`). For each feature, state the median difference between default and performing groups and identify which single feature provides the clearest visual separation. (3)
- (f) Plot a grouped bar chart of default rate by loan grade (A through G). Confirm whether the grade system is monotonically ordered. Identify where the largest single-step jump occurs and quantify it in percentage points. (2)
- (g) Calculate the default rate for each loan purpose and display as a horizontal bar chart, sorted descending. Identify the three highest-risk and two lowest-risk loan purposes. Quantify the ratio between the highest and lowest default rates. (2)
- (h) Identify the ten states with the highest default rates and display as a bar chart. Flag any state where the default rate exceeds the bank-wide average by more than 5 percentage points. (2)
- (i) Plot annual default rate from 2010 to 2024 as a line chart. Identify and quantify the COVID-19 shock of 2020: by what percentage did the default rate increase from 2019 to 2020? Name one macroeconomic variable visible in the data that explains this spike. (2)
- (j) Create a dual-axis line chart showing `rbi_repo_rate` and annual default rate on the same time axis. Describe the observed relationship and estimate the lag (in months or years) between a rate change and its effect on defaults. (2)
- (k) Plot a histogram with KDE overlay for `lgd_pct` on defaulted loans only. Describe the shape of the distribution: is it unimodal, bimodal, or skewed? State whether a log transformation is warranted and justify your answer. (1)
- (l) Produce a scatter plot of `cibil_score` vs. `lgd_pct` with a fitted regression line. Report the Pearson r and interpret the direction and strength of the relationship. (0)

Twelve engineered features are mandatory. For each feature you must: (i) state the formula, (ii) document the domain rationale in one to two sentences, (iii) verify the output with `.describe()`, and (iv) compute the Pearson correlation with `lgd_pct`.

- (a) Construct the following four repayment-burden features and for each report the correlation with `lgd_pct`: $\text{emi_to_income_ratio} = \text{installment_inr} \div (\text{annual_inc_inr} \div 12)$ $\text{loan_to_income_ratio} = \text{loan_amnt_inr} \div \text{annual_inc_inr}$ $\text{rate_spread_pct} = \text{int_rate_pct} - \text{rbi_repo_rate_pct}$ $\text{real_interest_rate} = \text{int_rate_pct} - \text{cpi_inflation_pct}$ Which single feature shows the strongest correlation with the target? (5)
- (b) Construct the following three bureau-behaviour features: $\text{credit_util_composite} = 0.5 \times \text{revol_util} + 0.3 \times \text{bc_util} + 0.2 \times \text{all_util}$ $\text{delinq_severity_score} = \text{delinq_2yrs} \times (1 + 1 \div \max(\text{mths_since_last_delinq}, 1))$ $\text{enq_velocity_score} = \text{num_enquiries_30d} \times 4 + \text{num_enquiries_90d}$ For `delinq_severity_score`, explain why recency weighting is preferable to a simple count. (5)
- (c) Construct the following three income and collateral features: $\text{income_stability_ratio} = \text{annual_inc_inr} \div (\text{emp_length_years} + 1)$ $\text{credit_depth_score} = \text{total_acc} \div (\text{credit_hist_years} + 1)$ $\text{collateral_coverage_ratio} = \text{collateral_value_inr} \div (\text{loan_amnt_inr} + 1)$ Explain in business terms what a high `collateral_coverage_ratio` implies for Loss Given Default. (5)
- (d) Apply $\log(1 + x)$ transformation to `annual_inc_inr` and `loan_amnt_inr` to create `log_annual_inc` and `log_loan_amnt`. Report the skewness of each variable before and after transformation. Explain why reducing skewness matters for OLS regression assumptions. (3)
- (e) Create a binary `covid_issue_year_flag` equal to 1 for loans issued in 2020. Present a grouped summary showing mean `lgd_pct` for `covid_issue_year_flag` = 0 vs. 1. Is the difference statistically significant? Run an independent-samples t-test and report the p-value. (2)

QUESTION 4 REGRESSION MODELLING & DIAGNOSTICS

[15 Marks]

All modelling must be performed on the defaulted-loans subset only (`df[df.loan_status == 1]`). Remove any column derived after default (`total_pymnt`, `recoveries`, `collection amounts`, `net_loss_inr`) before modelling to prevent data leakage.

- (a) Compute Variance Inflation Factors (VIF) for all candidate features. Present the full VIF table. Drop any feature with $\text{VIF} > 10$ and justify each removal decision. State the final list of features retained for modelling. (4)
- (b) Fit a baseline OLS model using `statsmodels`. Print the full model summary. Interpret: R^2 , Adjusted R^2 , F-statistic (with p-value), Durbin-Watson statistic, (6)

Jarque-Bera test result, and Condition Number. For the five largest-magnitude coefficients, provide a one-sentence business translation of each.

- (c) Fit Ridge, Lasso, and ElasticNet models using GridSearchCV with 5-fold cross-validation. For each: report the optimal alpha, cross-validated RMSE, and test-set R^2 . Present a side-by-side comparison table. List all features that Lasso reduced to exactly zero and comment on what their exclusion implies. (5)
- (d) Produce a 4-panel diagnostic figure: (i) Residuals vs. Fitted, (ii) QQ Plot of residuals, (iii) Scale-Location, (iv) Cook's Distance. For each panel explicitly state which regression assumption it tests and whether that assumption is satisfied or violated in your model. (10)

QUESTION 5 BUSINESS RECOMMENDATIONS

[10 Marks]

Translate your analytical findings into five board-ready recommendations. Each recommendation must include: (i) the specific data evidence or statistic that motivates it, (ii) a quantified business impact (in ₹ Crore or percentage points where possible), and (iii) a concrete implementation action.

- (a) Based on your EDA findings on CIBIL score distributions, EMI-to-income ratios, loan grades, and loan purpose default rates, formulate five distinct recommendations that the Credit Risk team could implement within the next financial year. Each recommendation must stand on its own with independent evidence and a separate action. (10)
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EXAMINER'S MARKING SHEET

Section	Max Marks	Marks Obtained
Q1 — Data Acquisition, Joining & Cleaning	30	
Q2 — Exploratory Data Analysis (12 Charts)	25	
Q3 — Feature Engineering	20	
Q4 — Regression Modelling & Diagnostics	15	
Q5 — Business Recommendations	10	
TOTAL	100	

Examiner's Name: _____

Signature: _____

Date of Evaluation: _____